

MH371 truth-blind cruise control

conditional_antenna_no_precomp_medium_bfo | 2 deterministic seeds | generated from run artifacts

Control result

MODEL FAMILY

conditional_antenna_no_precomp_medium_bfo

PARTICLES

12000 x 2 seeds

TERMINAL MEAN ERROR

2.7 to 37.5 NM

TERMINAL MASS WITHIN 100 NM

59.9 to 64.3%

TERMINAL ROOT ESS

34.1 to 35.4

BFO OBSERVATION SD

4.0 Hz

TRUTH-IMPLIED BFO BIAS

167.2 to 171.6 Hz

BTO EQUATION CLOSURE

3.076e-1 us max

Evidence and inference conditions

- Cruise only: 01:48 UTC initial state through the selected 06:48:33 UTC SATCOM observation.
- Later ACARS positions, trajectory, Mach, altitude, and headings were excluded from inference and loaded only here.
- ERA5 temperature and wind plus IGRF-14 east-positive magnetic declination drive every propagation step.
- Satellite oscillator and Perth GES AFC terms remain separate inputs; aircraft BFO bias is analytically estimated.
- Each map combines deterministic seed replicates with equal replicate weight.
- Conditional RxGain endpoint: no precompensation; event-level SD 1.72 dB; reference gain 12.16 dBic.
- Power rows use the selected event/channel aggregation; maximum absolute offset from its SATCOM arc is 73.922 s.

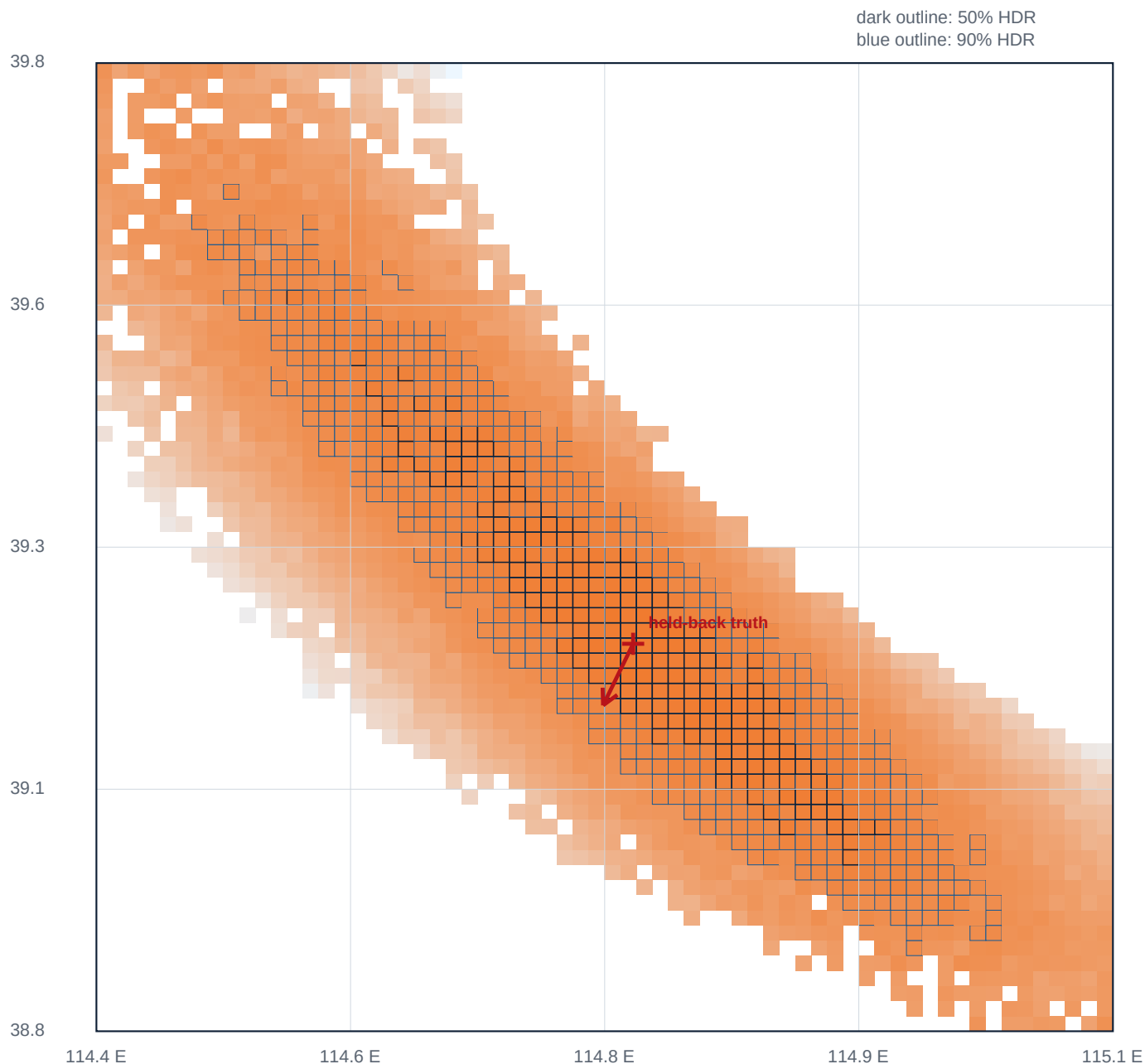
Material limitations

- held-back ACARS positions and trajectories were loaded only by this scorer
- posterior localization is conditional on the declared motion and BFO-noise family
- this known-flight control does not establish adequacy for the MH370 accident flight
- conditional antenna sensitivity only: RxGain validation did not justify unconditional core use or a model-family mixture weight
- the explicitly permitted first-pass gain surface is uncommissioned, lacks verified port/starboard handoff, and does not reproduce every workbook event gain
- the event/channel full-precompensation prediction is frozen; the antenna likelihood contributes only the modeled directional departure

The following 6 pages show posterior density, held-back position, and true-heading arrow at each selected SATCOM arc.

mh371_0200 SATCOM arc

2014-03-07T01:56:47.921Z | truth 39.2204 deg, 114.7725 deg | true heading 206.0 deg



Posterior and control metrics

MEAN POSITION ERROR

2.4 NM

MACH: POSTERIOR / TRUTH

0.783 / 0.759

GROUND SPEED: POSTERIOR / TRUTH

387 / 383 kt

TRUTH-IMPLIED BFO BIAS

167.2 Hz

POSTERIOR WITHIN 100 NM

100.0%

ALTITUDE: POSTERIOR / TRUTH

34289 / 27599 ft

LATENT BFO BIAS

166.9 Hz

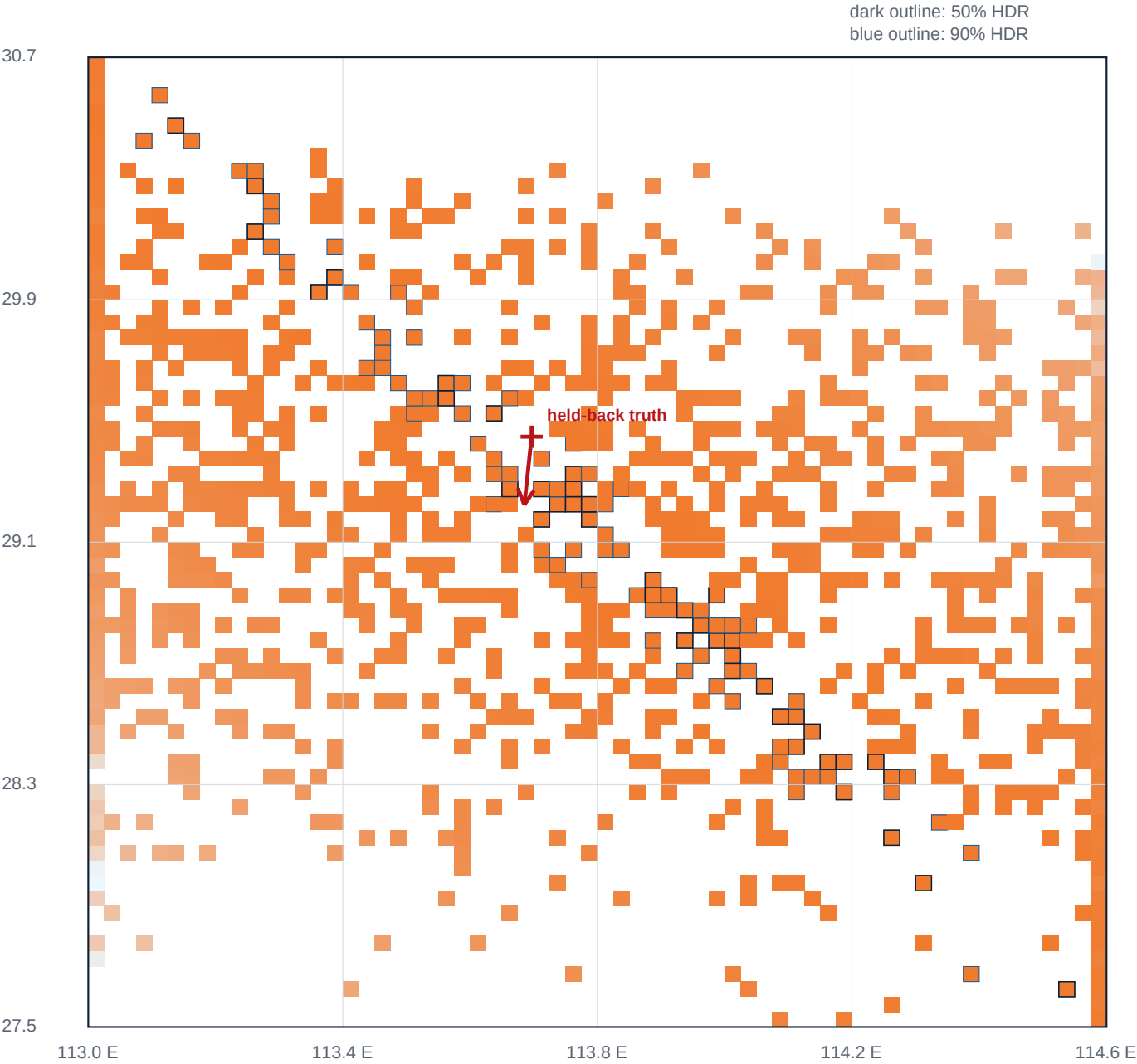
PARTICLE / ROOT ESS

7682 / 7682.1

Density combines equal-weight deterministic seed replicates. The red cross and arrow were loaded only during scoring.

mh371_0300 SATCOM arc

2014-03-07T03:20:22.428Z | truth 29.4730 deg, 113.6856 deg | true heading 186.1 deg



Posterior and control metrics

MEAN POSITION ERROR
22.7 NM

MACH: POSTERIOR / TRUTH
0.787 / 0.753

GROUND SPEED: POSTERIOR / TRUTH
434 / 452 kt

TRUTH-IMPLIED BFO BIAS
171.6 Hz

POSTERIOR WITHIN 100 NM
98.1%

ALTITUDE: POSTERIOR / TRUTH
35440 / 28368 ft

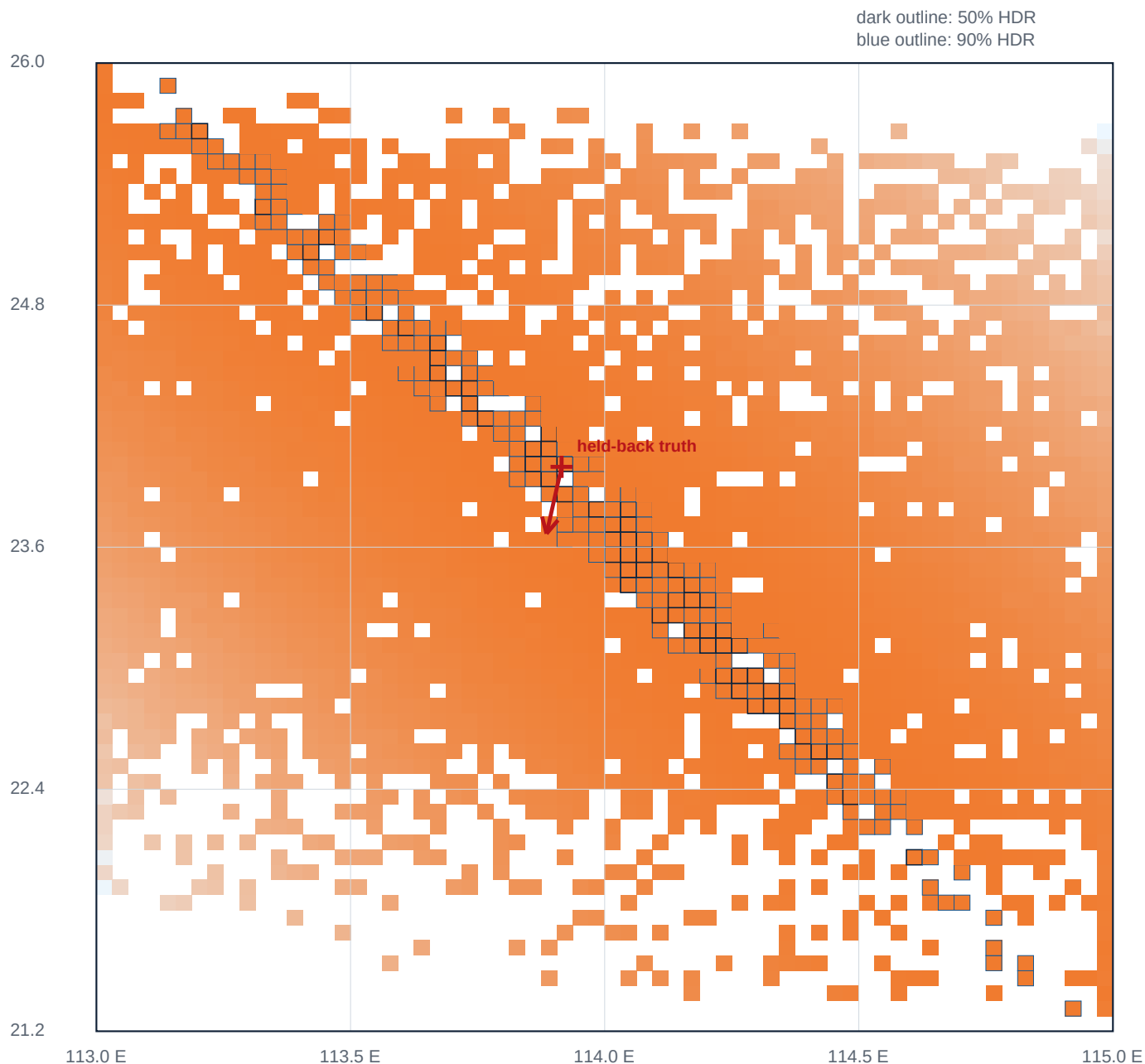
LATENT BFO BIAS
168.7 Hz

PARTICLE / ROOT ESS
72 / 71.7

Density combines equal-weight deterministic seed replicates. The red cross and arrow were loaded only during scoring.

mh371_0400 SATCOM arc

2014-03-07T04:04:57.916Z | truth 23.9533 deg, 113.8817 deg | true heading 192.7 deg



Posterior and control metrics

MEAN POSITION ERROR

20.4 NM

MACH: POSTERIOR / TRUTH

0.787 / 0.832

GROUND SPEED: POSTERIOR / TRUTH

441 / 455 kt

TRUTH-IMPLIED BFO BIAS

169.6 Hz

POSTERIOR WITHIN 100 NM

88.4%

ALTITUDE: POSTERIOR / TRUTH

35430 / 38100 ft

LATENT BFO BIAS

169.1 Hz

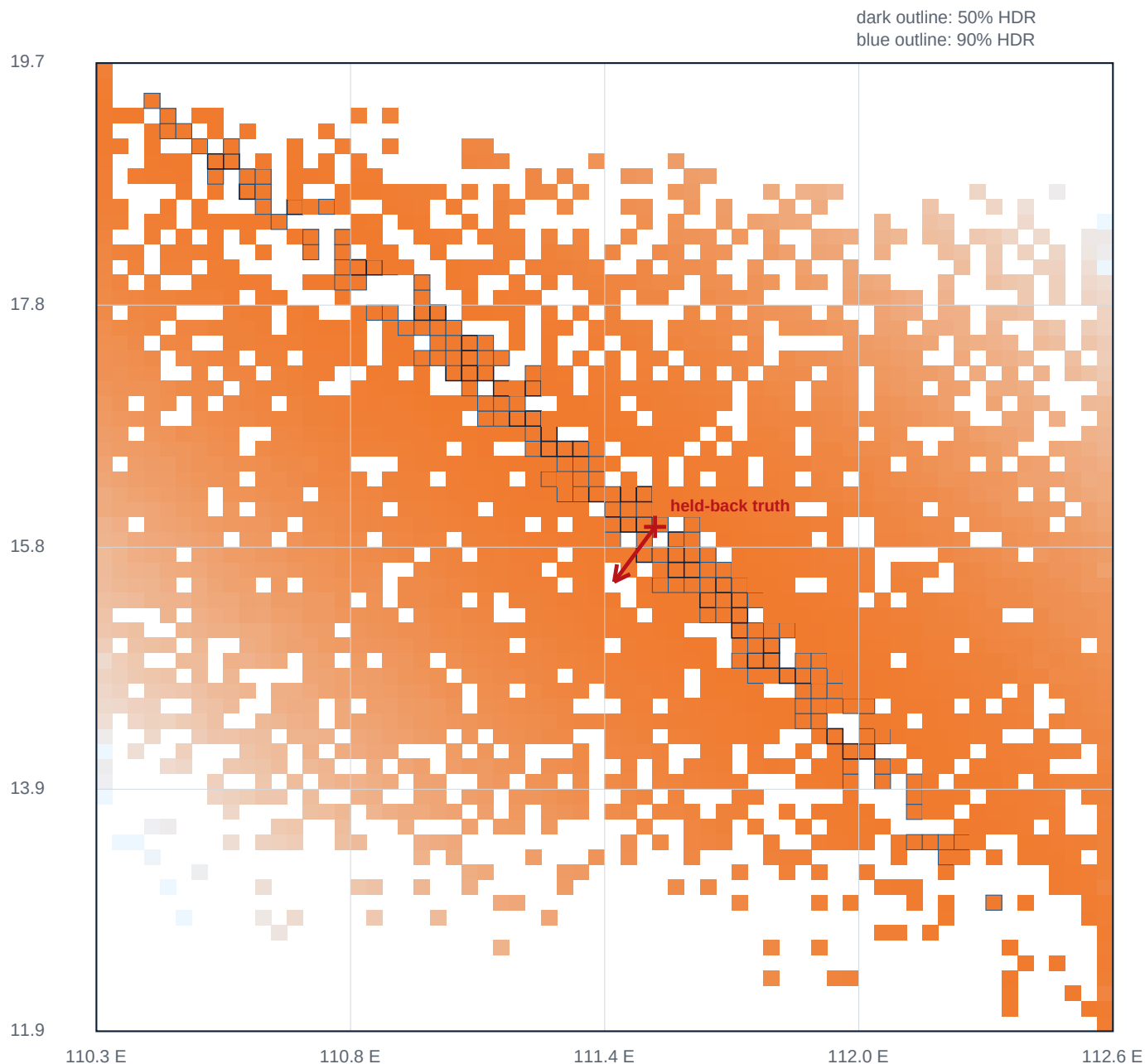
PARTICLE / ROOT ESS

293 / 58.9

Density combines equal-weight deterministic seed replicates. The red cross and arrow were loaded only during scoring.

mh371_0500 SATCOM arc

2014-03-07T05:11:37.411Z | truth 15.9956 deg, 111.5330 deg | true heading 216.7 deg



Posterior and control metrics

MEAN POSITION ERROR

20.5 NM

MACH: POSTERIOR / TRUTH

0.774 / 0.830

GROUND SPEED: POSTERIOR / TRUTH

442 / 462 kt

TRUTH-IMPLIED BFO BIAS

169.7 Hz

POSTERIOR WITHIN 100 NM

70.2%

ALTITUDE: POSTERIOR / TRUTH

35984 / 39998 ft

LATENT BFO BIAS

169.0 Hz

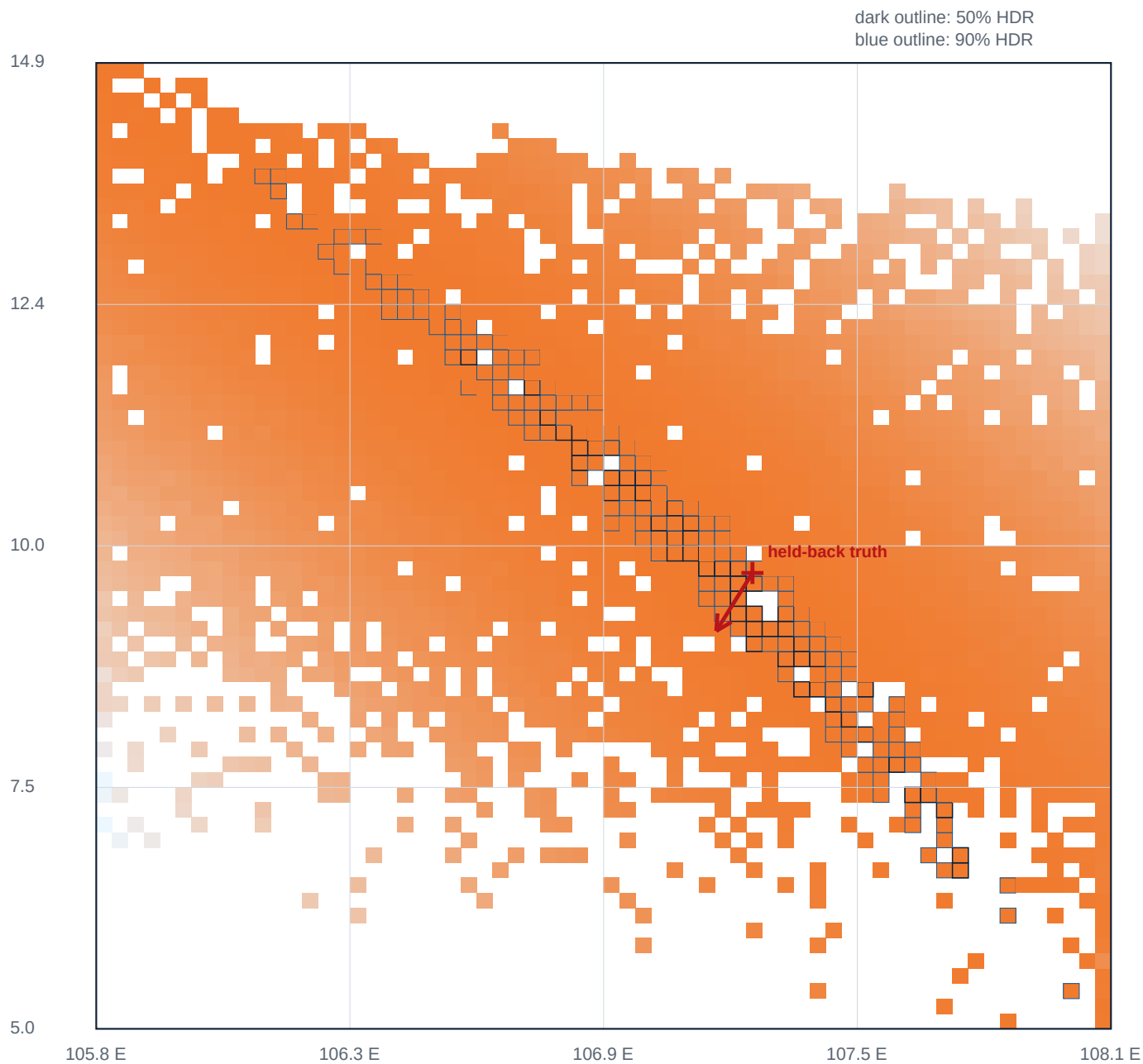
PARTICLE / ROOT ESS

166 / 43.3

Density combines equal-weight deterministic seed replicates. The red cross and arrow were loaded only during scoring.

mh371_0600 SATCOM arc

2014-03-07T06:09:44.409Z | truth 9.6835 deg, 107.2620 deg | true heading 212.0 deg



Posterior and control metrics

MEAN POSITION ERROR

18.4 NM

MACH: POSTERIOR / TRUTH

0.789 / 0.813

GROUND SPEED: POSTERIOR / TRUTH

470 / 474 kt

TRUTH-IMPLIED BFO BIAS

170.6 Hz

POSTERIOR WITHIN 100 NM

65.3%

ALTITUDE: POSTERIOR / TRUTH

34454 / 39997 ft

LATENT BFO BIAS

169.3 Hz

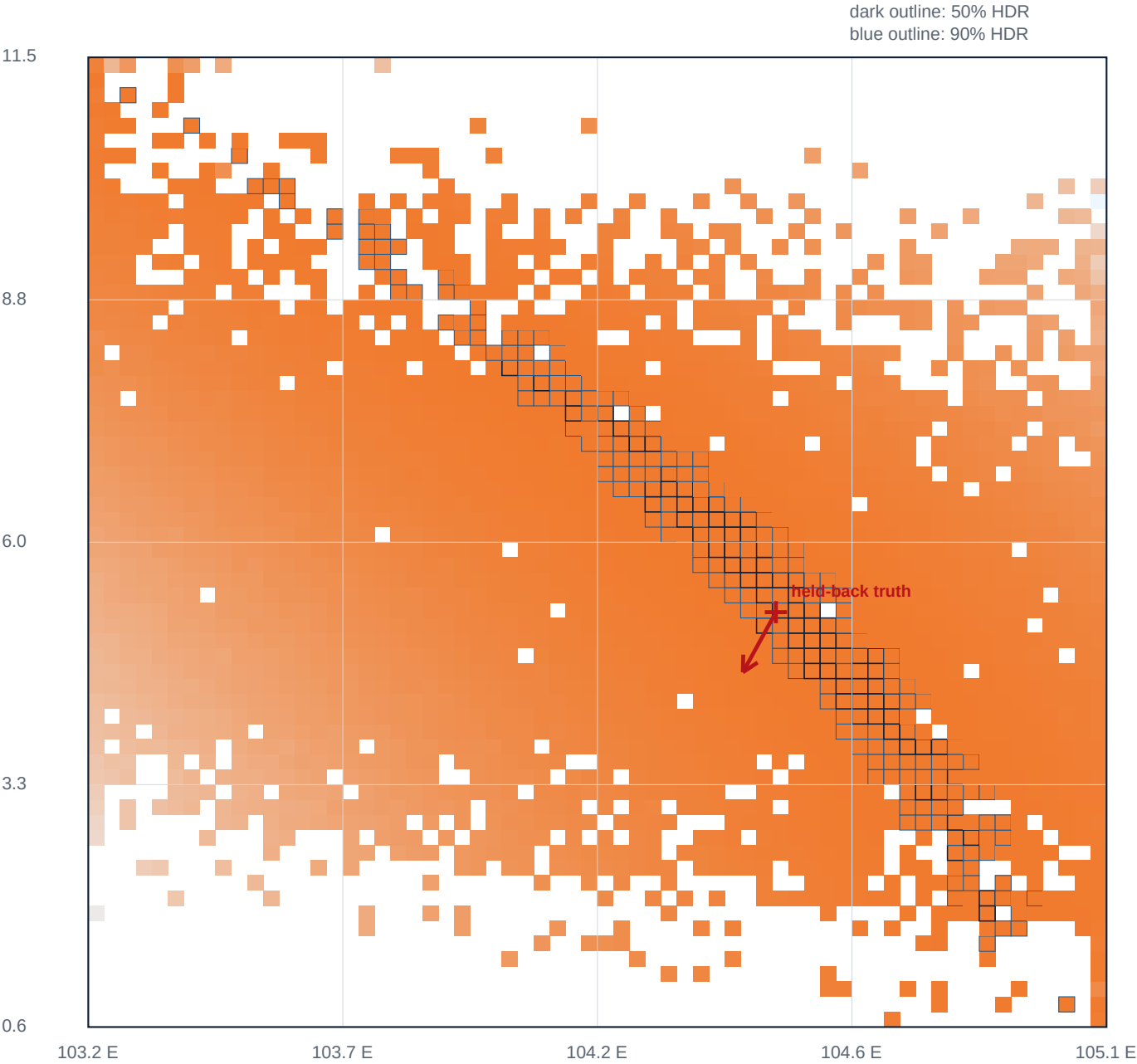
PARTICLE / ROOT ESS

208 / 37.2

Density combines equal-weight deterministic seed replicates. The red cross and arrow were loaded only during scoring.

mh371_0648 SATCOM arc

2014-03-07T06:48:33.907Z | truth 5.2504 deg, 104.4874 deg | true heading 208.8 deg



Posterior and control metrics

MEAN POSITION ERROR
20.1 NM

MACH: POSTERIOR / TRUTH
0.786 / 0.821

GROUND SPEED: POSTERIOR / TRUTH
475 / 483 kt

TRUTH-IMPLIED BFO BIAS
168.6 Hz

POSTERIOR WITHIN 100 NM
62.1%

ALTITUDE: POSTERIOR / TRUTH
34605 / 40000 ft

LATENT BFO BIAS
169.5 Hz

PARTICLE / ROOT ESS
477 / 34.7

Density combines equal-weight deterministic seed replicates. The red cross and arrow were loaded only during scoring.